

Industry Problem Task — Problem Statement

AI-Based Diabetes Diagnosis & Personalised Treatment Recommendation System

Course:	Artificial Intelligence (CSA17)	Course Outcome:	CO4 (BL3)
Assessment:	Tool 1 — Annexure A	Weightage:	50%
Questions:	4 (with sub-parts a/b/c)	Total Marks:	40
Duration:	1 Hour	Domain:	Healthcare AI/ML

Industry Scenario

A healthcare company has collected patient records — Age, Blood Pressure, Cholesterol, Glucose level, BMI, and Family History — with the goal of building a predictive model for the early diagnosis of diabetes. In addition to diagnosis, the company wants to train an intelligent agent that can recommend personalised treatment actions (**Diet / Exercise / Medication / Monitor**) based on how a patient's health state changes over time.

You have been hired as an **AI/ML Engineer** for this project. You are required to design, build, and evaluate a complete pipeline that spans inductive (supervised) learning for diagnosis and reinforcement learning for treatment recommendation, and to justify every design decision from both a technical and a clinical standpoint.

Task 1 — Data Preparation & Inductive Learning

(a) Approach & Feature Identification

Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features, with justification for each.

(b) Class Imbalance Handling

Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy — SMOTE, under-sampling, or class weights.

(c) Train/Test Split & Preprocessing

Split the dataset 70:30 and justify the split ratio for a medical use case. State the preprocessing steps (normalisation, encoding) applied and their impact on model quality.

Task 2 — Decision Tree for Diagnosis

(a) Tree Construction & Root Node Justification

Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluation & Clinical Risk of False Negatives

Evaluate the model by reporting Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them, with clinical justification.

(c) Post-Pruning

Apply post-pruning using cost-complexity pruning. Compare pre-pruning versus post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3 — Statistical Learning for Risk Stratification

(a) Statistical Model & Comparison

Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree, and discuss when a statistical model is preferable.

(b) Feature Importance Analysis

Perform feature importance analysis. Identify the top three predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4 — Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

Model the treatment recommendation problem as a Markov Decision Process (MDP). Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), the Reward function (health-improvement score), and Transition dynamics. Justify each component.

(b) Q-Learning Training

Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Policy Extraction & Ethics

Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations — patient safety, bias, and explainability.

Evaluation Rubric

Criteria	Weightage	Focus of Assessment
Understanding of Industry Problem	20%	Grasp of real-world problem and constraints
Application of Engineering Concepts	25%	Correct, effective use of AI/ML concepts
Solution Design	25%	Feasibility, innovation, industry alignment

Criteria	Weightage	Focus of Assessment
Use of Tools	15%	Effective use of tools, technologies, real data
Analysis	10%	Depth of analysis and result validation
Professionalism	5%	Documentation quality and presentation

Deliverables Expected

- Problem Statement document (this document)
- Solution document with complete written answers to all sub-questions
- Python source code implementing the full pipeline
- Output visualisation(s) (charts / decision tree / dashboard)
- A professional project report summarising methodology and results
- A README file describing how to run the project